

EXPERIMENT 5

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A PYTHON PROGRAM TO IMPLEMENT MUTILAYER PERCEPTION WITH BACK PROPAGATION

AIM:

*TO IMPLEMENT A PYTHON PROGRAM WITH MUTILAYER PERCEPTION
WITH BACK PROPAGATION*

CODE:

```
import pandas as pd
import numpy as np
bnotes = pd.read_csv('/content/BankNote_Authentication.csv')
bnotes.head(10)

x = bnotes.drop('class',axis=1)
y = bnotes['class']
print(x.head(2))
print(y.head(2))

from sklearn.model_selection import train_test_split
#train_test ratio = 0.2
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2)
from sklearn.neural_network import MLPClassifier
# activation function : relu
mlp = MLPClassifier(max_iter=500,activation='relu')
mlp.fit(x_train,y_train)
MLPClassifier(max_iter=500)
pred = mlp.predict(x_test)
print(pred)
```

```
from sklearn.metrics import classification_report, confusion_matrix
confusion_matrix(y_test,pred)
print(classification_report(y_test,pred))
```

```
# activation function : logistic
mlp = MLPClassifier(max_iter=500,activation='logistic')
mlp.fit(x_train,y_train)
```

```
MLPClassifier(activation='logistic', max_iter=500)
pred = mlp.predict(x_test)
print(pred)
```

```
from sklearn.metrics import classification_report, confusion_matrix
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

```
mlp = MLPClassifier(max_iter=500,activation='tanh')
mlp.fit(x_train,y_train)
pred = mlp.predict(x_test)
print(pred)
```

```
from sklearn.metrics import classification_report, confusion_matrix
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

```
# activation function : identity
mlp = MLPClassifier(max_iter=500,activation='identity')
mlp.fit(x_train,y_train)
MLPClassifier(activation='identity', max_iter=500)
pred = mlp.predict(x_test)
print(pred)
```

```
from sklearn.metrics import classification_report, confusion_matrix
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

```
#train_test ratio = 0.3
```

```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.3)
```

```
from sklearn.neural_network import MLPClassifier
```

```
# activation function : relu
```

```
mlp = MLPClassifier(max_iter=500,activation='relu')
```

```
mlp.fit(x_train,y_train)
```

```
MLPClassifier(max_iter=500)
```

```
pred = mlp.predict(x_test)
```

```
print(pred)
```

```
from sklearn.metrics import classification_report,confusion_matrix
```

```
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

```
# activation function : logistic
```

```
mlp = MLPClassifier(max_iter=500,activation='logistic')
```

```
mlp.fit(x_train,y_train)
```

```
MLPClassifier(max_iter=500,activation='logistic')
```

```
pred = mlp.predict(x_test)
```

```
print(pred)
```

```
MLPClassifier(max_iter=500,activation='tanh')
```

```
# activation function : tanh
```

```
mlp = MLPClassifier(max_iter=500,activation='tanh')
```

```
mlp.fit(x_train,y_train)
```

```
pred = mlp.predict(x_test)
```

```
print(pred)
```

```
from sklearn.metrics import classification_report,confusion_matrix
```

```
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

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# activation function : identity
mlp = MLPClassifier(max_iter=500,activation='identity')
mlp.fit(x_train,y_train)
MLPClassifier(max_iter=500,activation='identity')
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```
from sklearn.metrics import classification_report,confusion_matrix
confusion_matrix(y_test,pred)
```

```
print(classification_report(y_test,pred))
```

OUTPUT:

```
IDLE Shell 3.12.3
File Edit Shell Debug Options Window Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr 9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/itzdi/OneDrive/Documents/ML_Codes/Exp_5.py
      variance  skewness  kurtosis  entropy
0      3.6216      8.6661     -2.8073    -0.44699
1      4.5459      8.1674     -2.4586    -1.46210
0      0
1      0
Name: class, dtype: int64
[0 1 1 0 1 0 1 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 1 1 0 1 0 1 0
 0 0 0 0 0 0 0 1 0 0 1 1 0 0 1 1 0 0 1 1 0 0 0 0 1 1 0 1 0 0 0 0 1 0 0 0 1 0
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 1 0 0 0 0 1 0 1 1 0 0 1 1 0 1 0]
      precision    recall  f1-score   support

         0         1.00      1.00      1.00        142
         1         1.00      1.00      1.00        133

 accuracy          1.00
 macro avg         1.00      1.00      1.00        275
 weighted avg      1.00      1.00      1.00        275
```

```
[0 1 1 0 1 0 1 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 1 0 1 0 1 0
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1 0 0 0 0 1 0 1 1 0 0 1 1 0 1 0]
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	142
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accuracy			1.00	275
macro avg	1.00	1.00	1.00	275
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[0 1 1 0 1 0 1 1 1 0 0 0 1 0 0 0 1 0 0 0 1 1 0 1 0 0 0 0 0 1 1 1 0 1 0 1 0
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1 0 0 0 0 1 0 1 1 0 0 1 1 0 1 0]
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[0 1 0 1 0 0 1 1 1 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 1 0 0 1 1 0 1 0
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1 0 0 1 0 1 0 0 1 0 0 1 1 0 1 0 0 1 1 1 1 1 0 0 0 1 0 1 1 1 0 0 0 0 0 0
1 1 1 0 0]
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	221
1	1.00	1.00	1.00	191
accuracy			1.00	412
macro avg	1.00	1.00	1.00	412
weighted avg	1.00	1.00	1.00	412

```
[0 1 0 1 0 0 1 1 1 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 1 0 0 1 1 0 1 0
1 0 0 1 1 0 0 0 1 1 0 1 1 0 1 1 1 1 1 1 0 0 0 1 1 0 0 0 1 1 0 0 0 0 1 1
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1 0 0 1 0 1 0 0 1 0 0 1 1 0 1 0 0 1 1 1 1 1 0 0 0 1 0 1 1 1 0 0 0 0 0 0
1 1 1 0 0]
```

```
[0 1 0 1 0 0 1 1 1 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 1 0 0 1 1 0 1 0
1 0 0 1 1 0 0 0 1 1 0 1 1 0 1 1 1 1 1 1 0 0 0 1 1 0 0 0 1 1 0 0 0 0 1 1
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1 0 0 1 0 1 0 0 1 0 0 1 1 0 1 0 0 1 1 1 1 1 0 0 0 1 0 1 1 1 0 0 0 0 0 0
1 1 1 0 0]
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	221
1	1.00	0.99	1.00	191
accuracy			1.00	412
macro avg	1.00	1.00	1.00	412
weighted avg	1.00	1.00	1.00	412

```

[0 1 0 1 0 1 1 1 1 0 0 1 0 0 0 1 0 1 0 0 0 0 0 0 1 1 1 0 0 1 0 0 1 1 0 1 0
 1 0 0 1 1 0 0 0 1 1 0 1 1 0 1 1 1 1 1 1 1 0 0 0 1 1 0 0 0 1 1 0 0 0 0 1 1
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 1 0 0 1 0 1 0 0 1 0 0 1 1 0 1 0 0 1 1 1 1 1 0 0 0 1 0 1 1 1 1 0 0 0 0 0 0
 1 1 1 0 0]
precision    recall  f1-score   support

     0        0.99        0.99        0.99        221
     1        0.98        0.98        0.98        191

 accuracy          0.99          412
 macro avg          0.99          412
weighted avg          0.99          412

```

RESULT:

A PYTHON PROGRAM TO IMPLEMENT MULTILAYER PERCEPTION WITH BACK PROPAGATION AS BEEN ANALYSED AND VERIFIED